PRACTICAL 6. Node.js and Express Fundamentals

PRACTICAL NO.6(A)

Aim:- Install Express and create a simple server.

Procedure:-

Part 1:- Create a New Express Project Using Command Prompt

Step 1:- Open Command Prompt

Open Command Prompt on the computer.

Step 2:- Create a new project folder

Enter the following command:-

mkdir express-practical1

Step 3:- Navigate to the project folder

cd express-practical1

Step 4:- Initialize the Node.js project

npm.cmd init -y

This creates a **package.json** file containing the basic configuration of the Node.js project.

Step 5: Install Express.js

npm.cmd install express

Express.js and its required dependencies will be installed in the project.

After successful installation, the following folders/files are created:

- **node_modules –** Contains installed packages.

- **package-lock.json –** Stores the exact versions of installed packages.
- **package.json –** Contains project information and dependencies.

Part 2: Create **server.js** File Using Visual Studio Code

Step 1:- Open Visual Studio Code

Launch Visual Studio Code.

Step 2:- Import/Open the project folder

1. Click **File** in the top menu.
2. Select **Open Folder...**
3. Navigate to:

C:\Users\user\express-practical1

4. Select the **express-practical1** folder.
5. Click **Select Folder**.

The project folder will now appear in the **Explorer** panel on the left side.

Step 3:- Create a new JavaScript file

1. In the Explorer panel, right-click the **express-practical1** folder.
2. Select **New File**.
3. Enter the file name → server.js
4. Press **Enter**.

The **server.js** file will be created inside the project folder.

Step 4:- Write the Express server code

Open **server.js** and write the following code:-

```
const express=require("express");

const app=express();

const PORT=3000;
```

```
app.get("/",(req,res)=>{
   res.send("Hello from Express!");
});

app.listen(PORT,()=>{
   console.log(`Server is running on
      http://localhost:${PORT}`);
});
```

Run the command (In CMD):-

→ node server.js

Open Browser and visit:-

→ http://localhost:3000

Browser Output:-

Hello from Express!

PRACTICAL NO.6(B)

Aim:- Display a message on the browser using the server response.

Code:-

```
const express=require("express");

const app=express();

const PORT=3000;

app.get("/",(req,res)=>{
   res.send("Hello! Welcome to my Express Server.");
});
```

```
app.listen(PORT,()=>{
   console.log(`Server is running on
      http://localhost:${PORT}`);
});
```

Open Browser and visit:-

→ http://localhost:3000

Browser Output:-

Hello! Welcome to my Express Server.

PRACTICAL NO.6(C)

Aim:- Create different routes in Express.

Code:-

```
const express=require("express");

const app=express();

const PORT=3000;

//Home Route
app.get("/",(req,res)=>{
   res.send("Welcome to the Home Page.");
});

//About Route
app.get("/about",(req,res)=>{
   res.send("This is About Page");
});

//Contact Route
app.get("/Contact",(req,res)=>{
   res.send("This is Contact Page");
});
```

```
//Products Route
app.get("/Products",(req,res)=>{
   res.send("This is Products Page");
});

app.listen(PORT,()=>{
   console.log(`Server is running on
      http://localhost:${PORT}`);
});
```

Open Browser and visit:-

→ http://localhost:3000

Browser Output:-

→ Visit http://localhost:3000

Welcome to the Home Page.

→ Visit http://localhost:3000/About

This is About Page

→ Visit http://localhost:3000/Contact

This is Contact Page

→ Visit http://localhost:300/Products

This is Products Page

PRACTICAL NO.6(D)

Aim:- Send a JSON response from the server.

Code:-

```
const express=require("express");

const app=express();

const PORT=3000;

//Home Route
app.get("/",(req,res)=>{
   res.send("Welcome to the Home Page.");
});

//About Route
app.get("/about",(req,res)=>{
   res.send("This is About Page");
});

//Contact Route
app.get("/Contact",(req,res)=>{
   res.send("This is Contact Page");
});

//Products Route
app.get("/Products",(req,res)=>{
   res.send("This is Products Page");
});

app.get("/api/user",(req,res)=>{
   res.json({
      id:1,
      name:"John Doe",
      email: "john@example.com",
      city: "Mumbai"
   });
});

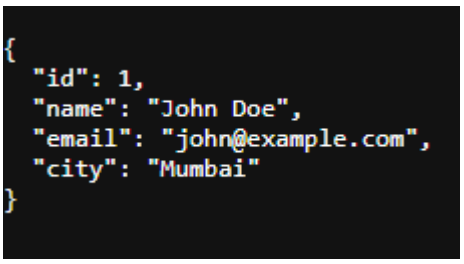
app.listen(PORT,()=>{
   console.log(`Server is running on
```

```
    http://localhost:${PORT}`);
});
```

Open Browser and Visit:-

→ http://localhost:3000/api/user

Browser Output:-



PRACTICAL NO.6(E)

Aim:- Test routes using browser or Postman.

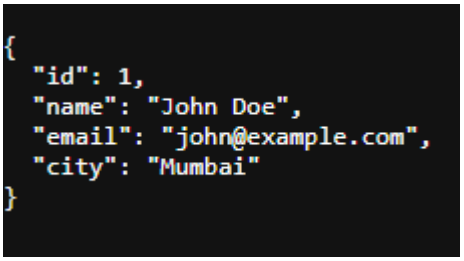
Code:-

In practical 6(c) we create, in practical 6(e) we test.

Route	Method	Expected Output
/	GET	Home Page
/about	GET	About Page
/contact	GET	Contact Page
/products	GET	Products Page
/api/user	GET	JSON user data

Routes directly in the browser.
For Eg:-
→ https://localhost:3000/api/user

Browser Output:-



PRACTICAL NO.6(F)

Aim:- Run the server and observe the output in the console.

Code:-

```
const express=require("express");

const app=express();

const PORT=3000;

//Home Route
app.get("/",(req,res)=>{
   console.log("Home Route accessed");
   res.send("Welcome to the Home Page.");
});

//About Route
app.get("/about",(req,res)=>{
   console.log("About route accessed");
   res.send("This is About Page");
});

//Contact Route
app.get("/Contact",(req,res)=>{
   console.log("Contact route accessed");
```

```
    res.send("This is Contact Page");
});

//Products Route
app.get("/Products",(req,res)=>{
    console.log("Products route accessed");
    res.send("This is Products Page");
});

app.get("/api/user",(req,res)=>{
    console.log("User API accessed");

    res.json({
        id:1,
        name:"John Doe",
        email: "john@example.com",
        city: "Mumbai"
    });
});

app.listen(PORT,()=>{
    console.log(`Server is running on
        http://localhost:${PORT}`);
});
```

Open terminal in VS Code & enter:-

→ node server.js

Open Browser and visit:-

→ http://localhost:3000
→ http://localhost:3000/api/user

VS Code Output:-

```
PS C:\Users\user\express-practical1> node server.js
Server is running on
        http://localhost:3000
Home Route accessed
User API accessed
User API accessed
```